

DAY 5 - Counters.

1) Seq: $0 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 3 \rightarrow 1 \rightarrow 0$
Using 3 bit T-FF

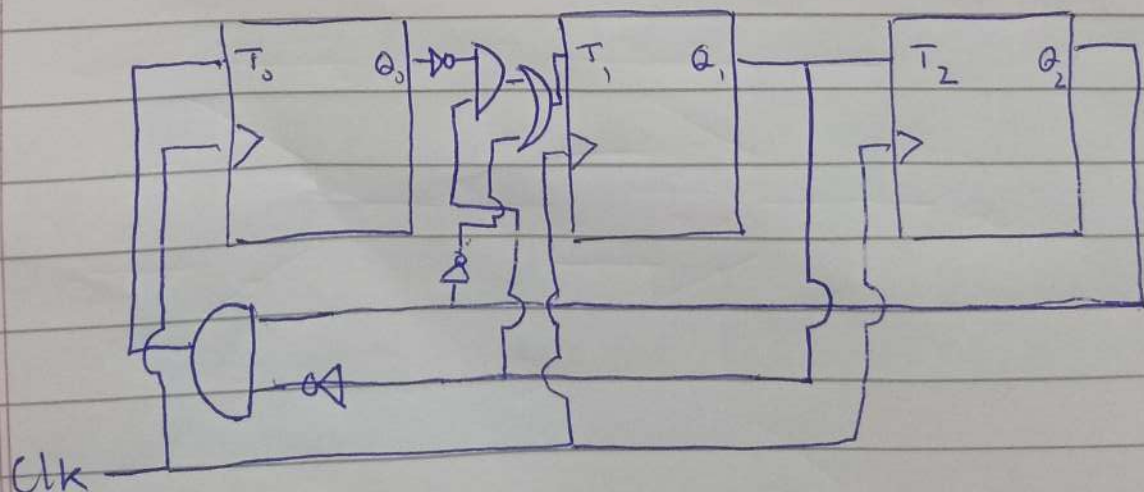
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0	0	1	0	0	1	0
0	0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	1	0
0	1	1	0	0	1	0	1	0
1	0	0	1	0	1	0	0	1
1	0	1	0	1	1	1	1	0
1	1	0	x	x	x	x	x	x
1	1	1	x	x	x	x	x	x

Using K map:

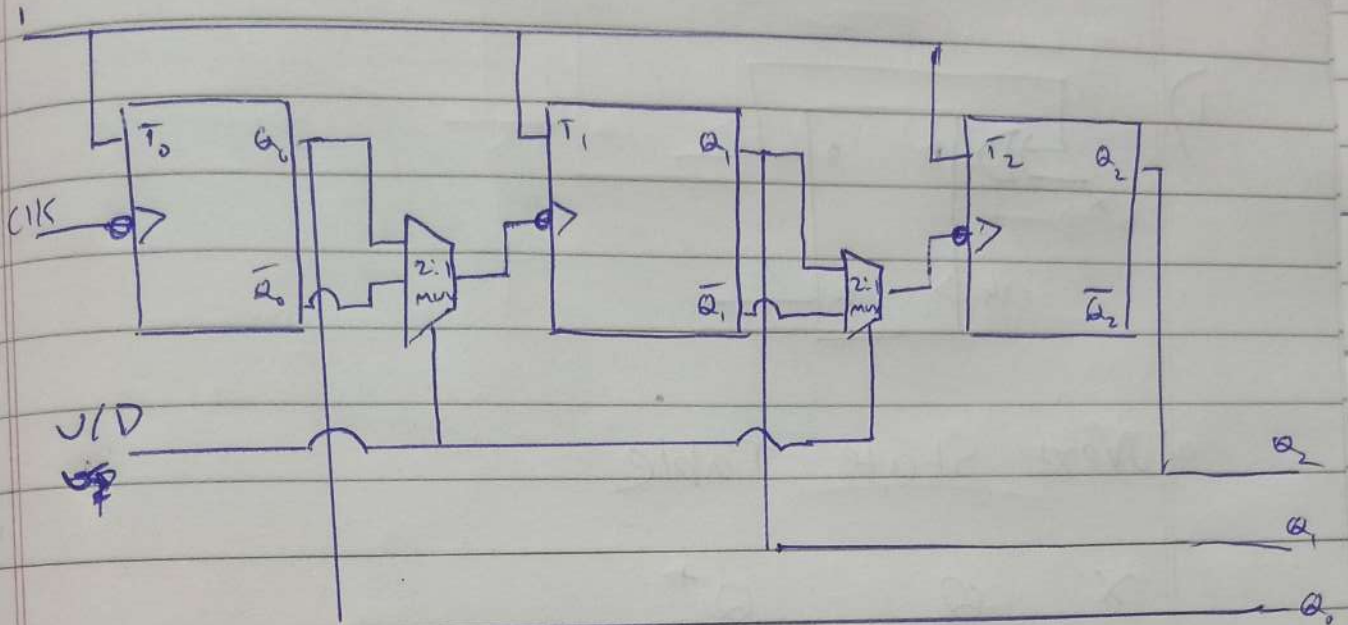
$$T_2 = Q_1$$

$$T_1 = \bar{Q}_2 + Q_1 \bar{Q}_0$$

$$T_0 = Q_2 \bar{Q}_1$$



2) edge Asynchronous 3 bit



$V/D \rightarrow 0 \rightarrow$ for upcount ($clk \rightarrow Q$)
 $V/D \rightarrow 1 \rightarrow$ for downcount ($clk \rightarrow \bar{Q}$)

$\downarrow$ CLK	Q_2	Q_1	Q_0	T_2	T_1	T_0	Q_2^+	Q_1^+	Q_0^+
0	0	0	0	0	0	0	0	0	1
1	0	0	1	0	1	1	0	1	0
2	0	1	0	1	0	1	1	1	1
3	0	1	1	1	1	1	0	0	0

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$